

SMARTSHEET × SAS/R • STUDY CP-101

Your tracker updates itself — and Smartsheet sends the alert

A nightly SAS/R job reads the CP-101 Program Tracker, updates status, % complete, and milestone dates from the validated data, cascades a moved date, attaches the latest status PDF — and a Smartsheet automated workflow emails the team. No AI. Operations data only, never PHI or a reported clinical number.

Illustrative mockups. Smartsheet is a tracker, not a system of record; validated systems own the authoritative data.

Scheduler

nightly • Task Scheduler / cron



SAS / R

PROC HTTP / httr2 → API



Smartsheet

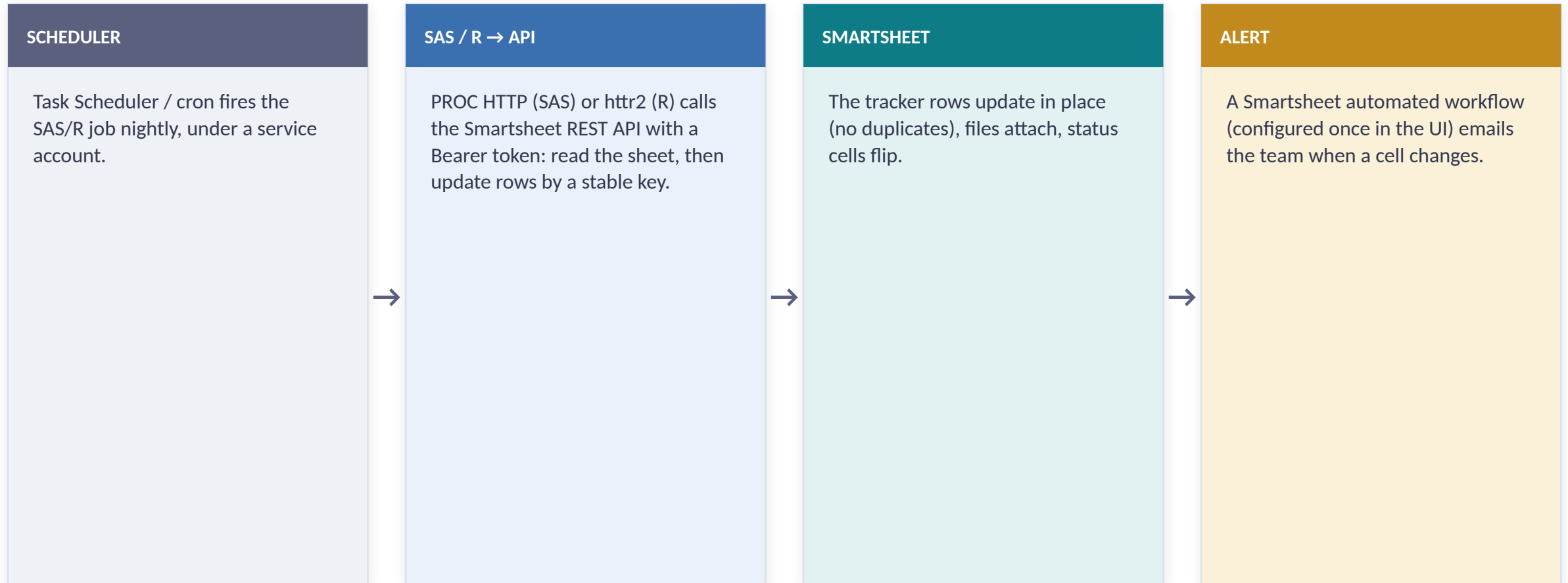
tracker rows updated by key



Smartsheet alert

a workflow emails the team

Scheduled SAS/R → the Smartsheet API → your tracker → an alert



The job reads the tracker — rows keyed by a Task ID

Smartsheet — CP-101 Program Tracker

Task ID	Task	Status	% Done	Owner	Due
CP101-ENR	Enrollment vs plan	On Track	—	PM	ongoing
CP101-ADPC	ADPC spec — finalize	On Track	60%	A. Patel	15 Jun
CP101-CUT	Database soft-lock / data cut	On Track	—	DM	17 Jun
CP101-TLF	Dry-run TLFs	On Track	10%	J. Kim	24 Jun
CP101-DSMB	DSMB pack	On Track	0%	Biostat	30 Jun

The job pulls the sheet (GET /sheets/{id}) and maps each row by its Task ID — so it updates the right rows in place, never appending duplicates.

1

Read first, key by Task ID. Idempotency is the whole game: update the existing row, never blind-append — so re-running the job never duplicates a thing.

Status, % complete & dates updated from the validated data

Smartsheet — CP-101 Program Tracker (after the nightly run)

Task ID	Task	Status	% Done	Owner	Due	
CP101-ENR	Enrollment vs plan — 76% of plan	At Risk	—	PM	ongoing	UPDATED
CP101-ADPC	ADPC spec — finalize	On Track	85%	A. Patel	15 Jun	UPDATED
CP101-CUT	Database soft-lock / data cut	Watch	—	DM	19 Jun	UPDATED
CP101-TLF	Dry-run TLFs	On Track	10%	J. Kim	26 Jun	UPDATED
CP101-DSMB	DSMB pack	On Track	0%	Biostat	30 Jun	

2

PUT /sheets/{id}/rows updates the cells by row id: enrollment flips to At Risk, ADPC jumps to 85%, the cut date moves to 19 Jun — straight from the validated study data.

The data cut moved 17 → 19 Jun — dependents follow

⋮ SAS / R — the cascade (deterministic)

```
/* the confirmed change */
data cut: 17 Jun -> 19 Jun (+2 days)

/* shift the dependent milestones */
soft-lock    22 -> 24 Jun
dry-run TLFs 24 -> 26 Jun
DSMB pack    30 Jun (holds)

/* one PUT updates the affected rows */
%ss_upsert(sheet=&tracker, keycol=TaskID,
            data=cascade_rows);
```

⋮ Smartsheet — affected rows

Task	Due (was)	Due (now)	
Data cut	17 Jun	19 Jun	UPDATED
Soft-lock	22 Jun	24 Jun	UPDATED
Dry-run TLFs	24 Jun	26 Jun	UPDATED
DSMB pack	30 Jun	30 Jun	

Dependent dates are computed in SAS/R and pushed in one update — the timeline stays consistent without anyone re-typing it.

Attach the status PDF — and let Smartsheet send the alert

Smartsheet — summary row

CP-101 — weekly status

 CP101_status_2026-06-16.pdf (attached by the job)

POST /sheets/{id}/rows/{rowId}/attachments — the latest one-page status PDF lands on the summary row, so the current report is always one click away in the tracker.

Outlook — from Smartsheet Automation

 **CP-101 Program Tracker — a task is now At Risk**

Smartsheet • Automated workflow • to: PM, Biostat lead

Row **CP101-ENR — Enrollment vs plan** changed to **At Risk** (76% of plan).

Open the tracker to review. — Sent by the Smartsheet workflow your team configured, triggered by tonight's update.

Four guardrails for an unattended SAS/R → Smartsheet job

Idempotent by key

Update existing rows by a stable Task ID — never blind-append. Re-running the job (or a retry) never creates duplicate rows.

Ops-only — no PHI

Push only timelines, status, % complete, milestone dates, ownership. NEVER PHI, participant-level data, unblinded info, or reported clinical numbers — a column allowlist enforces it.

Token kept secret + fail-loud

The API token lives in a permissioned credentials file / env var, never in code. A missing/expired token or an API error stops loud and emails the backup — never a silent no-op.

Rate-limit aware

Handle HTTP 429 with exponential backoff and batch updates; the tracker stays consistent even under throttling.

The boundary: Smartsheet is a cloud tracker — operational data only. The validated systems own the authoritative numbers; Smartsheet shows the status.

One macro to push status; Smartsheet does the rest.

Maximum ease

A thin SAS/R macro library wraps the Smartsheet API — one call pushes a status, attaches a report, or cascades a date. Schedule it once; the tracker maintains itself.

SAS/R does the data; Smartsheet does the alert

Deterministic API calls from your validated programs; the rich notifications are Smartsheet workflows you configure once in the UI.

Governed by design

Idempotent updates, an ops-only column allowlist (no PHI), a secret token, and fail-loud error handling.

Fits the family

The no-AI companion to the SAS/R trial-monitoring automation — same scheduler, same discipline, a different target.